

SAGE

Structured AI-Guided Education

Your Guide to Using AI Responsibly

Six steps that make your learning visible and defensible

Developed by Dr Mahmoud Elkhodr & Professor Ergun Gide

Central Queensland University · sage-framework.com

What is SAGE?

SAGE is a six-step process for using generative AI tools in your coursework and assessments. It does not determine whether you may use GenAI — your assessment instructions and institutional policy set that rule. It explains how to use GenAI so that the thinking and learning in your work are genuinely yours.

AI tools are fast and fluent. They can produce a convincing-looking answer to almost any question in seconds. That is the problem. Fluent does not mean correct, and correct for one context does not mean appropriate for yours. SAGE gives you a repeatable process to catch what AI gets wrong, add what it misses, and document the reasoning that shows you actually understand your work.

The short version: generate a draft with AI → check it against real sources → fix what is wrong → ask AI to criticise the developing work and judge that criticism → reflect on what you learned → demonstrate that the learning is yours. Each step leaves a record of your thinking.

Before You Start: Find Your Anchors

Every step in SAGE involves checking AI output against something authoritative. That authoritative source is called an anchor. Your lecturer will usually provide your anchors — a marking rubric, a required reading, a professional standard, or a regulatory framework.

If your lecturer has not specified anchors, you need to find them yourself. This is a real professional skill. In your career, you will always need to know what the authoritative reference is for your field before you can evaluate whether AI output is reliable.

Your discipline	Where to find your anchors
Cybersecurity	NIST Cybersecurity Framework, ISO 27001, Privacy Act 1988, ASD Essential Eight, ACSC guidelines
Health / Nursing	Clinical practice guidelines, NDIS Practice Standards, NSW Health policies, AHPRA codes, drug reference compendia
Business / Management	Relevant legislation (Fair Work Act, Corporations Act), company policies, ASX governance codes, industry codes of conduct
Engineering	Australian Standards (AS series), Safe Work Australia codes, engineering codes of practice, environmental regulations
Law	Relevant statute, case law, Legal Aid guidelines, court rules, professional conduct rules
Education	Australian Curriculum (ACARA), state syllabus documents, UDL guidelines, school/institution policy
Not sure?	Start with your unit reading list. Find the key textbook or foundational paper for your topic. Ask your librarian for the peak professional body in your field.

💡 How to verify a source is authoritative

Ask: Is this published by a government body, a peak professional association, a regulatory agency, or a peer-reviewed journal? If yes, it can be an anchor. A blog post, a Wikipedia article, or another AI-generated summary cannot.

Step 1 1 Generate

Start with your thinking — not the AI's.

Before you type a single word into an AI tool, spend five to ten minutes writing down what you already know about the task. It does not need to be polished. Bullet points, rough notes, or a badly worded draft all count. The point is that your thinking comes first.

Then build your prior knowledge into your prompt. Give the AI context about the task, the constraints, and the standards that apply. A prompt that includes what you already know will produce far more useful output than a blank question.

💡 Why this matters

AI produces generic output in response to generic prompts. When you include your own context — your field, your constraints, your standards — the AI has something to work with. You also immediately see where the AI diverges from what you already knew, which is exactly what you need for Step 2.

Use your lecturer's base prompt if one has been provided. If not, use this structure:

EXAMPLE PROMPT

I am a [year level] [discipline] student. The task is: [describe the task in your own words]. The relevant standard or framework for this context is [anchor]. My initial thinking is: [paste your notes or draft here]. Based on this context, [specific request].

💡 What if you have no prior knowledge at all?

Write down what you understand the task to be asking, which terms you recognise, and where your understanding breaks down. Even a statement of what you do not yet know is a valid starting point — and it is far more useful to include in your prompt than nothing.

Step 2 2 Evaluate

Check the AI's work against something real.

Read the AI output carefully. It will probably look confident and well-structured. That does not mean it is right. AI tools frequently miss context-specific requirements, omit regulatory details, produce plausible-sounding but incorrect facts, and apply generic advice where specialist knowledge is needed.

Compare the AI output against your anchors point by point. For each claim the AI makes, ask:

- Is this accurate according to my anchor?
- Is this complete, or has the AI missed something important?
- Is this appropriate for my specific context, or is it generic advice that does not fit here?
- Has the AI omitted any regulatory, ethical, cultural, or stakeholder requirements?

Take notes on everything you find. These notes become your decision log in Step 3.

Common AI failure patterns to look for

AI frequently: omits enforcement mechanisms; misapplies standards from a different jurisdiction; gives generic best practice rather than context-specific requirements; misses exception handling and edge cases; and ignores cultural, accessibility, or stakeholder-specific factors.

3 Step 3 **Refine**

Fix it, own it, document every decision.

Now revise the AI output based on what you found in Step 2. For every significant element, make a deliberate decision and record it in your log. There are only three options:

ACCEPT The AI output is accurate and appropriate for your context. You can explain why, citing your anchor.

MODIFY The AI output is partially right but needs adjustment. Record exactly what you changed and why, citing the specific source or reasoning.

REJECT The AI output is wrong, misleading, or inappropriate for your context. Record what you removed and what the correct position is, with a reference.

Use the table below to record your decisions. Your log is the core evidence that the thinking in your work is yours. Keep it.

AI Output — paste or summarise the claim	Decision	Justification — cite a specific source, standard, or reason (50–75 words)
<i>Example: "AI recommended optional encryption for stored data"</i>	<i>Reject</i>	<i>Encryption must be mandatory for health records under Privacy Act APP 11.1. "Reasonable steps" threshold is not met by optional controls. Source: OAIC guidance on APP 11.</i>

 **How many decisions should you log?**

Aim for at least one logged decision per significant paragraph or section of the AI output. A surface-level assessment with only two or three entries does not demonstrate genuine critical engagement. Six to ten decisions for a standard-length task is a reasonable target.

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Step 4

AI Critic

Audit the critic

You have made your decisions and documented your reasoning. Now ask GenAI to act as a critic, examiner, reviewer, auditor or devil's advocate.

Write a new prompt that tells the AI to act as a professional reviewer matched to the anchor you used in Step 2. Give it your decision log and ask it to find weaknesses in your reasoning. Then evaluate its criticism rather than automatically accepting it.

EXAMPLE PROMPT

You are a [professional role — e.g., senior cybersecurity auditor / clinical reviewer / compliance officer]. I have made the following decisions when evaluating AI-generated content against [your anchor]. Review my log and identify: where I may have accepted something I should have questioned; where a justification is weak or missing a citation; and where my reasoning may not hold in this professional context. [Paste your decision log here]

The AI will flag concerns about your decisions. Your job is to evaluate those flags — not just accept them. Record your adjudication in the table below.

AI flag — what the AI criticised	Your decision	Your reason — why you accept or reject this flag (30–50 words)

The AI Critic activity is a single bounded pass. You do not keep going back and forth with the AI. One set of flags, one set of adjudications, documented. You are the decision-maker — the AI is the challenger.

When to reject an AI flag

Reject the flag when: the AI is applying a generic principle that does not fit your specific context; the AI does not know about a stakeholder constraint or local requirement you do know; or the AI is repeating a concern you already addressed in your log. Always record why you are rejecting it.

5 **Reflect**

Step 5

What did you actually learn?

Look back over your decision log and your AI Critic adjudications. This is where you turn the experience into learning that will carry forward beyond this task.

Answer the four questions below. Your reflection should be specific — not general observations about AI, but concrete observations about what this particular AI tool did in this particular disciplinary context.

Reflection question	Your answer
<i>What did the AI consistently get wrong in this task?</i>	
<i>Where did your own knowledge prove essential — something the AI could not have known?</i>	
<i>What is the right "anchor" for this kind of task in a professional context after graduation?</i>	
<i>What would you do differently next time?</i>	

The habit SAGE builds is not "check AI with these specific sources." It is "identify the right anchor for any context, then hold AI accountable against it." In your career, the anchors will change with every role and every task. The judgement stays the same.

Steps 1–5 prepare you for Defend

Your decision log and reflection are your preparation. Defend is not a test of memory. It asks you to explain, apply and adapt the capability represented in your final work.

Step 6

6 **Defend***Show that the learning is yours.*

In some assessments, your submitted work will be followed by a supervised activity in which you explain, apply, modify or reproduce the capability you developed. SAGE calls this Defend.

Defend is not a test of whether you can memorise your submission word for word. It is an opportunity to demonstrate that you understand the work, can explain the decisions represented in it, and can apply the same capability when the question, context, data or requirements change.

Your educator will tell you whether Defend forms part of your assessment, what format will be used, how long it will take and how it will be marked.

What might Defend involve?

- an interactive oral assessment or viva;
- structured individual questioning after a presentation;
- a code walkthrough or live debugging task;
- applying a statistical method to an unseen dataset;
- explaining a design, calculation, model or technical process;
- modifying an artefact when a requirement or constraint changes;
- responding to an unseen professional scenario;
- completing a short supervised practical task;
- explaining which contributions came from GenAI and which decisions were yours; and
- identifying errors, assumptions, limitations or alternative approaches.

What must you be able to do?

1. **Explain it** — Describe the purpose, reasoning and meaning of every major part in your own words.
2. **Trace it** — Identify what GenAI contributed, what you contributed and where your judgement changed the work.
3. **Justify it** — Explain why you accepted, modified or rejected important AI suggestions using evidence, standards, sources or disciplinary reasoning.
4. **Test it** — Check claims, calculations, references, code, data, assumptions, edge cases and limitations.
5. **Adapt it** — Change the work when the requirements, audience, scenario, data or constraints change.
6. **Reproduce the capability** — Complete a similar task, or part of it, without asking GenAI to reproduce the original answer.
7. **Be honest about limits** — If you do not know an answer, explain what you would check or how you would approach the problem rather than guessing or bluffing.

CQUniversity example

At CQUniversity, a Tier 2 medium-security guided-assured assessment combines a lower-weighted GenAI-supported preparatory component with a higher-weighted mandatory interactive component. Every student must explain, justify, critique and extend the work in real time. See sage-framework.com/cqu-guided-assured.html#tier2.

Preserve an authentic GenAI record

Where your assessment requires an AI log, decision log, chat export or declaration:

- preserve the complete and authentic evidence required by the assessment;
- do not fabricate, selectively alter or misrepresent AI interactions;
- retain the required records until the assessment and any review period are complete;
- record meaningful accept, modify and reject decisions;
- retain the sources, standards and evidence used to check AI output;
- protect personal, confidential and restricted information; and
- follow your educator's storage and submission instructions.

Your log supports transparency and helps you prepare, but the log alone does not prove that you developed the capability. You must also be able to explain, apply and adapt the learning represented in the final work.

Prepare before submission

- ☐ Can I explain every major part without reading from the AI output?
- ☐ Can I identify an important error, limitation or assumption?
- ☐ Can I justify my choices using an authoritative anchor?
- ☐ Can I explain the trade-offs behind my decisions?
- ☐ Can I modify the work for a changed requirement?
- ☐ Can I complete a similar task without reproducing the same AI conversation?
- ☐ Can I distinguish my contribution from the AI contribution?

Practise before you Defend

Complete GenAI-101, particularly Lesson 7, “Use AI as a Critical Coach”, and the Transfer Test activities in the later lessons. These activities help you practise explaining, checking, changing and reproducing AI-assisted work under different conditions. See sage-framework.com/genai-101.html.

ATTRIBUTION & LICENCE

Developed by Dr Mahmoud Elkhodr and Professor Ergun Gide, Central Queensland University. Validated across 800+ students. Listed in the TEQSA Generative AI Knowledge Hub. CC BY 4.0 — free to use and adapt.

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